ASSESSMENT TOOL 2-DEBATE

COURSE CODE: ITA14 **COURSE NAME: Ethical Hacking**

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Date	15/10/2026

COURSE OUTCOME COVERED

CO1: Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

Assessment Tool Weightage: 40%

Duration: 5 min Total Marks: 50

DEBATE TOPIC

Ethics of Ethical Hacking , Legal System, Proper and Ethical Disclosure

Code of Conduct:

I Rakshana AR (1921260027). (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ETHICAL HACKING — DEBATE ANSWER SHEET

Topic: Which is Best — Ethical Hacking by Humans or AI?

1. Introduction

Ethical hacking means testing computer systems with proper permission to find security weaknesses before criminals exploit them. Both human ethical hackers and Artificial Intelligence (AI) can help protect systems, but they have different strengths. Humans provide judgment, responsibility and context, while AI can analyse large amounts of data quickly. In my view, the best approach is human-led ethical hacking supported by AI.

2. Ethical Hacking by Humans

- Human hackers can understand the business situation, user behaviour and the possible real-world impact of avulnerability.
- They can make ethical decisions, ask for clarification and communicate responsibly with the organisation.
- A responsible human tester must follow the agreed scope, protect confidential information and disclose findingsonly to authorised people.
- However, humans can make mistakes, misuse their access or intentionally leak confidential information. Trustalone is not enough; safeguards are necessary.

3. Ethical Hacking by AI

- AI can process logs and large datasets quickly, identify unusual patterns and help prioritise possible vulnerabilities.
- It can support repetitive testing and operate consistently when correctly configured and supervised.
- AI may produce false positives, miss context or make errors. Its results need verification, and access to sensitivesystems must be controlled.
- AI does not automatically guarantee privacy or security: people still design, deploy, configure and oversee AI tools.

4. My Main Argument: What if an Ethical Hacker Changes Companies?

Suppose a person works in Company A's ethical hacking team and learns its network design, vulnerabilities, security controls and confidential reports. Later, that person joins Company B. What guarantees that the person will not share Company A's secrets? There is no absolute guarantee based on trust alone. This is an insider-risk problem, and it shows why organisations need both ethical professionals and strong security controls.

- Least-privilege access: give each employee only the access required for their role.
- Confidentiality agreements and clear policies: define how sensitive information must be handled, including afteremployment ends.
- Logging and monitoring: record access to sensitive files and investigate suspicious activity lawfully.
- Data minimisation and secure storage: share only the information needed, encrypt sensitive reports and restrictdownloads or transfers where appropriate.
- Responsible disclosure: report vulnerabilities only to authorised contacts through approved channels.
- Offboarding: promptly revoke access and recover company devices when an employee leaves.

Important point: A contract can create legal consequences for leaking secrets, but it cannot guarantee that a leak will never happen. Organisations reduce risk through layered safeguards, audits and accountability.

5. Human vs AI — Quick Comparison

Factor	Human ethical hacker	AI-assisted testing
Context and judgment	Strong at interpreting context and consequences	May miss context or make errors
Confidentiality risk	Insider misuse is possible	
Accountability	A person and organisation can be held responsible under applicable rules	
Best use	Planning, judgment, validation and reporting	Pattern detection, prioritisation and repetitive analysis

6. Legal and Ethical Responsibilities

Ethical hacking must be authorised, limited to the agreed scope and carried out in accordance with applicable law and company policy. Findings should be documented accurately and reported to authorised people without exposing unnecessary personal or confidential data. Unauthorised access, misuse of data or disclosure of trade secrets may lead to disciplinary or legal consequences, depending on the facts and applicable law.

7. Five-Minute Debate Speech

Good morning respected faculty and my fellow students. Today, our topic is: Which is better, ethical hacking by humans or by AI? I believe that human-led ethical hacking supported by AI is the strongest approach.

AI is fast. It can scan large amounts of data, identify unusual patterns and help find possible vulnerabilities. But speed is not the same as understanding. AI can make mistakes, miss context and produce results that need to be checked. Human ethical hackers can interpret the situation, assess the impact and communicate findings responsibly.

My main concern is confidentiality. Imagine an ethical hacker works for Company A and learns its vulnerabilities, network design and security reports. Then the hacker leaves and joins Company B. What assurance does Company A have that its confidential information will not be leaked? There is no perfect guarantee simply because someone is called an ethical hacker. People can misuse access, so companies need least-privilege access, confidentiality agreements, monitoring, secure report storage and immediate access removal when employees leave.

But AI is not automatically risk-free either. If sensitive company data is sent to an unapproved AI tool, that can also create a confidentiality risk. AI systems need controlled access, careful data handling and human supervision.

Therefore, I do not think the answer is to trust humans blindly or to replace them completely with AI. Humans should remain responsible for decisions, authorisation and ethical judgment, while AI assists with speed and analysis. The best cybersecurity comes from skilled people, useful AI tools, strong legal rules and clear accountability working together. Thank you.

8. Possible Questions and Short Answers

Q: Can AI completely replace ethical hackers?
A: No. AI can assist with analysis, but human oversight is needed for context, validation, authorisation and accountability.

Q: Are human hackers always more trustworthy?
A: No. People can misuse access, so organisations must use controls and audits rather than relying on trust alone.

Q: Is AI completely safe with company data?
A: No. Sensitive data should only be used with approved tools and suitable privacy and security controls.

Q: What is responsible disclosure?
A: Reporting a discovered vulnerability confidentially to the authorised organisation or contact so it can be fixed safely.

Q: What is your final conclusion?
A: Human-led ethical hacking with AI assistance, supported by legal compliance and strong confidentiality controls.

Key line to remember

“Trust is important, but trust alone is not a security control. Human judgment, AI assistance, confidentiality safeguards and accountability must work together.”